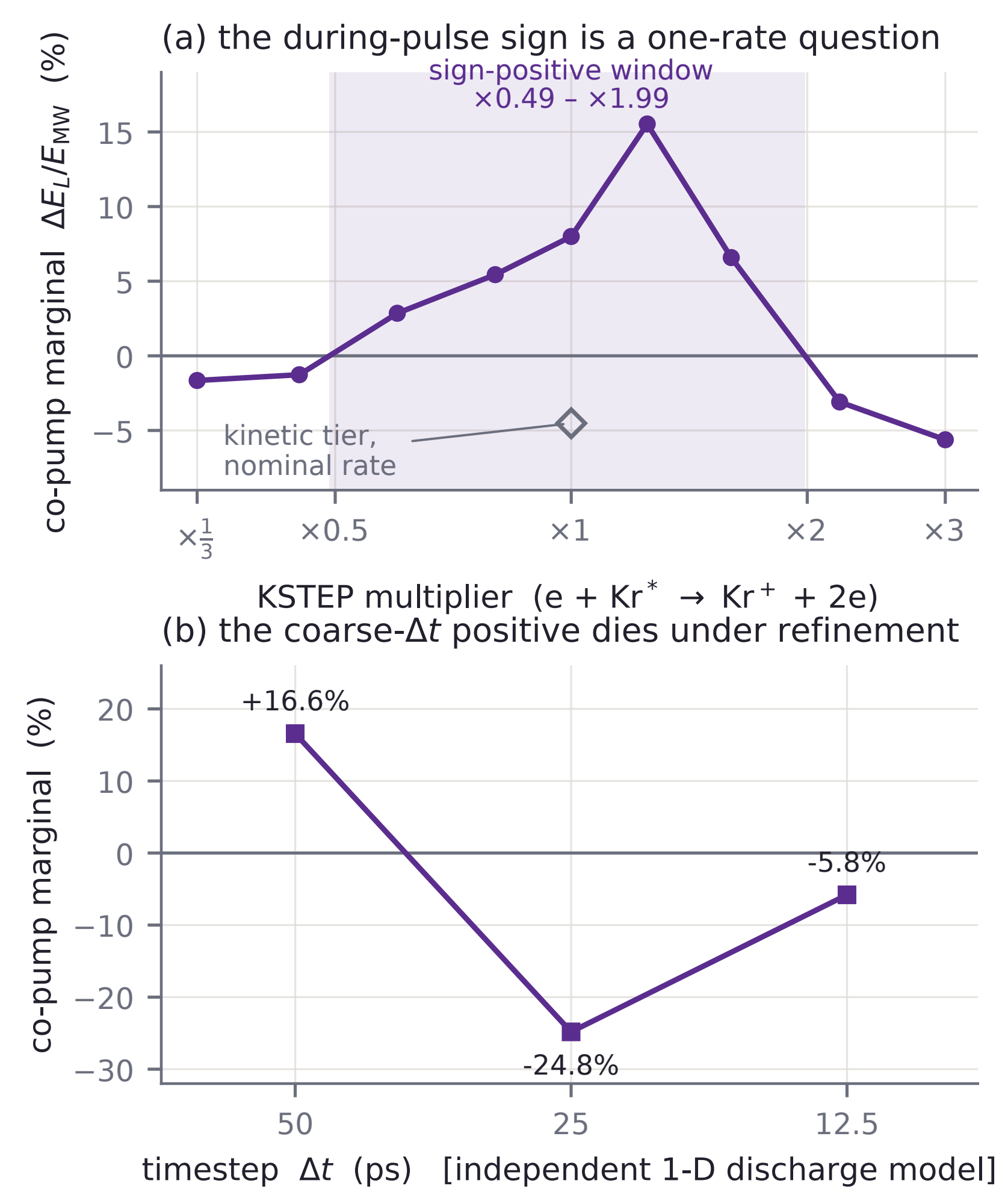
(a) the during-pulse sign is a one-rate question
sign-positive window
×0.49 – ×1.99
co-pump marginal $\Delta E_L/E_{MW}$ (%)
15
10
5
0
−5
kinetic tier,
nominal rate
×1/3
×0.5
×1
×2
×3
KSTEP multiplier (e + Kr* → Kr+ + 2e)
(b) the coarse-Δt positive dies under refinement
co-pump marginal (%)
20
10
0
−10
−20
−30
+16.6%
-24.8%
-5.8%
50
25
12.5
timestep Δt (ps) [independent 1-D discharge model]